MUHAMMAD SALMAN AL FARISI

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EDUCATION

National University of Singapore Aug 2023 – May 2027

Bachelor of Science (Hons.) in Data Science and Analytics

- Cumulative GPA: 4.64 / 5.00
- Double Major in Computer Science (Focus Area: Artificial Intelligence); Minor in Quantitative Finance
- Teaching Assistant for CS2040 Data Structures and Algorithms (4.8/5.0), IT1244 Artificial Intelligence & Machine Learning, and DSS5105 Data Science Projects in Practice

WORK EXPERIENCES

ShopBack, Automation Engineer Intern Jan 2026 – May 2026

- Built **PRISM**, a scalable end-to-end Quality Control automation pipeline reconciling merchant-reported and internal ShopBack data, now live in Australia and architected for rollout across 11 additional markets; features scheduled QC runs, email-triggered execution, and custom web crawling to verify that merchant contracts match their live page listings
- Developed **VoiceTrace**, an LLM-powered operations intelligence system for cashback dispute tickets that automates transaction retrieval, generates resolution insights, enables bulk ticket categorization for merchant-wide incident handling, and delivers real-time merchant health monitoring with automated Slack alerts across markets

Astra International, Data Scientist Intern July 2025 – Sep 2025

- Evaluated statistical and LLM-based evaluation frameworks to establish a baseline methodology for Astra's LLM-evaluator product
- Trained CPO price forecasting models for the Indonesian market, combining econometric and machine learning approaches with multi-source features to minimize prediction error while preserving interpretability
- Built a product-substitutability recommendation system for the spare-parts retail sector, covering market analysis, product bundling, and intelligent substitution strategies to improve sales and customer retention

National Health Group, Data Analyst Intern May 2025 – July 2025

- Designed a data pipeline that integrated and standardized questionnaire submissions from multiple formats into a single structured dataset for statistical analysis
- Automated personalized healthcare report generation, translating patient survey responses and domain knowledge into ready-to-use clinical summaries
- Applied multiple LLMs to convert unstructured triage and psychotherapy notes into structured JSON datasets categorized by psychological framework for downstream analysis

Mitra Taxindo Consulting, Web Developer (Freelance) Jan 2025 – May 2025

- Built and deployed a production bilingual (EN/ID) marketing website (mtaxindo.com) for a Jakarta tax-consulting firm using React 18, Vite, and React Router on Vercel, with a custom React-Context i18n layer, browser-language detection, and localStorage persistence
- Engineered SEO and performance optimizations — route-level code splitting, a per-route meta/Open-Graph hook, JSON-LD structured data, sitemap/robots, scroll-reveal animations, and a serverless Web3Forms contact pipeline with accessible feedback states

VetBuddy, Machine Learning Scientist Nov 2024 – Jan 2025

- Partnered with Mandai to deliver an AI-based medication-prescription web solution for zoos and veterinary clinics in Singapore and India
- Implemented Retrieval-Augmented Generation (RAG) with GPT-4, an Azure vector database, and Llama models to recommend accurate medication prescriptions from historical data and medication references

National University of Singapore, Teaching Assistant Jan 2025 – Dec 2025

- Taught weekly Data Structures & Algorithms and Machine Learning / AI classes for ~50 students, including preparing materials, answering forum questions, and guiding complex problem-solving
- Developed an automated grading pipeline for ~100 students, eliminating manual grading overhead and ensuring consistent, reproducible scoring at scale

AWARDS & ACHIEVEMENTS

- Awarded top student in Data Structures & Algorithms among 616 students for achieving highest cumulative score in the course

PROJECTS

Step-aware Large Language Model for Algorithm Navigation (saLLMan) - *PyTorch, RoPE, SwiGLU, FlashAttention, GRPO, BPE*

- Built a decoder-only LLM from scratch in PyTorch, hand-coding modern LLaMA-class internals (RoPE, SwiGLU, RMSNorm, Pre-LN, FlashAttention, KV-cache, weight tying); a controlled ablation at matched compute beat a vanilla GPT baseline by 9.9% validation perplexity (64.1 vs 71.2)
- Pretrained a 97M-parameter model on 2.2B tokens of Python on a single 8 GB RTX 3060 Ti, pairing a custom 16k byte-level BPE with aggressive memory engineering (bf16, fused AdamW, gradient checkpointing/accumulation to 131k-token batches, memory-mapped loading) to reach val loss 1.15 over ~3.6 epochs with no overfitting
- Supervised fine-tuned on Codeforces solutions with response-masked loss under a structured problem→reasoning→code format; hands-on data auditing drove two redesigns (swapping 15k-token traces for concise editorials, extending context 512→2048) for leaner training to val loss 1.38

- Implemented Group Relative Policy Optimization (GRPO) from scratch (DeepSeek-R1 recipe) with a critic-free two-model design that fits 8 GB, training the policy against a code-execution reward that runs generated Python against held-out test cases in a sandboxed subprocess
- Discovered the RL policy was reward hacking — printing constant outputs to farm partial test-pass credit — and engineered a baseline-subtracting reward that neutralized it; the hardened reward exposed that nearly all the apparent gain was gamed (genuinely-solvable problems fell from ~25% to under 2%), a rigorous negative result pinpointing model scale as the true ceiling

Document-Grounded Multimodal RAG – *RAG, FAISS, BM25, Cross-Encoder Re-ranking, PyMuPDF, Flask*

- Built a multi-format document Q&A system with a two-stage retrieval pipeline, hybrid FAISS + BM25 with Reciprocal Rank Fusion, followed by a cross-encoder re-ranker (ms-marco-MiniLM), evaluated on the FinanceBench benchmark
- Implemented multimodal ingestion (PDF, DOCX, TXT, Markdown) with image extraction and captioning via GPT-4o-mini vision, converting charts and diagrams into searchable text chunks
- Developed a Flask web interface and Jupyter evaluation pipeline with LLM-as-judge scoring across faithfulness, relevance, citation quality, and correctness, including adversarial nonsense-question testing to measure abstention rate

Interpretability of Sentiment Control in Seq2Seq Transformers – *Transformer, Ablation Study, Custom Attention Mechanism*

- Trained and compared N-gram, LSTM, and Seq2Seq Transformer architectures for sentiment-conditioned text generation, with the Transformer achieving 97.8% sentiment accuracy evaluated via pre-trained classifier
- Designed and executed 4 interpretability experiments to investigate how sentiment metadata is captured during autoregressive generation, identifying a 2-step mechanism of initial attention priming followed by self-attention structural context
- Proved sentiment information is strictly localized to the sentiment encoder token via perturbation and ablation experiments, and uncovered a learned positivity bias through mid-generation sentiment switching tests across 200 samples per condition

Twitter Bias Detector (Browser Extension) – *Natural Language Processing, Bayesian Optimization, Browser Extension Development*

- Hyperparameter-tuned Transformer-based model (DistilBERT) using low rank adaptations (LoRA) + Bayesian Optimization (OPTUNA) to reduce computing time and improve efficiency and increase overall metrics (accuracy, precision, recall & f1-score) by 20% from baseline models
- Designed a full-stack system integrating FastAPI backend, Hugging Face inference, and Chrome local storage (IndexedDB) to ensure data privacy and real-time user analytics

Tweets (Text & Image) Classification – *Natural Language Processing, Image classification, Text Sentiment Analysis*

- Collected and processed data by web scraping from Twitter to assign appropriate emojis to text using machine learning techniques, leveraging the Twemoji dataset containing tweet IDs, attached images and their corresponding emojis
- Implemented NLP preprocessing (Stop words removal, stemming, lemmatization) and tried several word embedding models (Word2Vec, Spacy, and BERT)
- Optimized and evaluated multiple ML models (*Logistic* Regression, CatBoost, LightGBM, SVM, and Neural Networks)

Stock Price Prediction – *Time Series Forecasting, Machine Learning Modelling, Deep Learning*

- Forecasted S&P 500 index closing prices by integrating historical prices (high, low, open, close) and company details alongside news sentiment analysis
- Hyperparameter-tuned and trained tree-based ensemble models (Random Forest, XGBoost, CatBoost) and recurrent neural networks architecture (RNN, GRU, LSTM)

COMPETITIONS

NVIDIA Nemotron Model Reasoning Challenge – *PyTorch, LoRA / PEFT, Nemotron-Nano (Mamba-Transformer MoE)*

- Fine-tuned NVIDIA's Nemotron-Nano (30B-total / 3B-active hybrid Mamba-Transformer MoE) with a rank-32 LoRA adapter on six families of symbolic reasoning puzzles, lifting the official grader score from a 0.53 baseline to 0.58
- Built a fully metric-compliant SFT pipeline (exact grader prompt format + \boxed{} target) trained only on puzzles whose reconstructed answers pass the grader's own verify(), balanced to 1,200 chain-of-thought examples per category

SKILLS

Programming Languages: Python, SQL, R, Java, C,
Machine Learning & Deep Learning: Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, HuggingFace (Transformers / Datasets / Tokenizers), SHAP, PEFT, LoRA Fine-Tuning, Mamba / State-Space Models (SSM), LangChain, Mixed-Precision (bf16) Training, Gradient Checkpointing & Accumulation, AdamW, Cosine LR Scheduling, GPU Memory Optimization
NLP & Generative AI: RAG, FAISS, BM25, Cross-Encoder Re-ranking, Seq2Seq Transformers, LLM-as-Judge, SFT
Databases: PostgreSQL, MySQL, IndexedDB
Website Development: ReactJS, Flask, FastAPI, Streamlit, Git & GitHub, Docker
Data Visualization: ggplot2 (R), Matplotlib (Python), PowerBI